

Battle of the Branches:

A comparison of Phylogeny-building tools, FASTTREE and IQ-TREE

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[Video Tutorial](#)

Primary Tools and Datasets

[FASTTREE, IQ-TREE](#)

Xylocopa sonorina (RefSeq: [GCF_050948175.1](#)),

Andrena cerasifolii (RefSeq: [GCF_050908995.1](#)),

Halictus rubicundus (RefSeq: [GCF_050948215.1](#)),

Colletes latitarsis (RefSeq: [GCF_051014445.1](#)),

Bombus vancouverensis nearcticus (RefSeq: [GCF_051014615.1](#)),

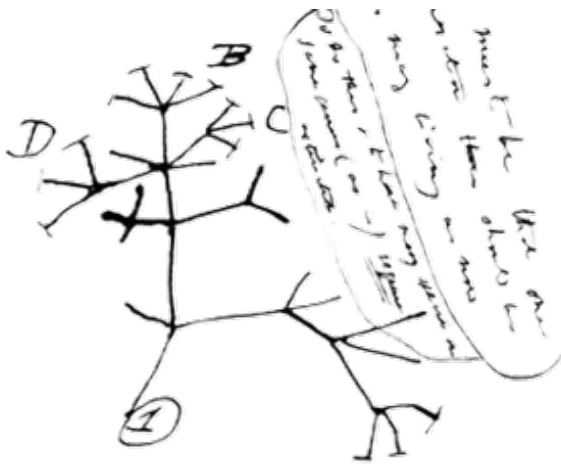
Solenopsis invicta (RefSeq: [GCF_016802725.1](#))

USER MANUAL

Purpose

The goal of this manual is to show users two different bioinformatic approaches for constructing phylogenies, a widely applied technique for elucidating evolutionary relationships between species or genes. Two tools from the Galaxy website, FASTTREE and IQ-Tree, will be discussed.

Background



The study of comparative evolution hinges on the knowledge of evolutionary relationships between groups. Phylogenetic trees are the most common way to represent these relationships due to their relative ease of interpretation. Branches are labeled as a given group, while nodes represent the last common ancestor of the groups. The tree originating from a single root demonstrates the idea that all species

are derived from a single common ancestor. See *Darwin's phylogeny [above]* (Thomas & Sherwin, 2009). Determining how the phylogeny should be constructed, however, is an entirely different issue. Historically, phylogenies have been built based on morphological characteristics. However, as DNA sequencing and bioinformatic analysis has become more accessible, phylogeny construction has relied increasingly on mapping these molecular relationships.

Bees were chosen as the subject of this manual, seeing how the success of this clade is so intimately linked with that of humans. Understanding the evolution of different bee species has practical implications for apiculture, as well as providing a better understanding of the insect's biogeographical history (Tihelka et al., 2020).

Bioinformatic Solution

As aforementioned, phylogenies may be constructed in a variety of ways. Using genetic data is widely considered to be the most reliable approach, as the genome reveals differences between species that morphology may not (Zou & Zhang, 2016). Consider convergent evolution- the concept that a phenotype evolved multiple times due to similar selective pressures. Convergent traits do *not* represent shared ancestry. For example, an echidna resembles a hedgehog or even a porcupine, but in reality, its closest living relative is the platypus (Zhou et al. 2021). Inversely,

morphological differences between species may be misleading in terms of organism's evolutionary relatedness, or lack thereof. For example, the closest living relative to the rock hyrax is the elephant (see *image [right]*) (Rock Hyrax 2016). This fact may not be captured by a morphological phylogeny depending on which trait is looked at.



Galaxy is a website designed for novice bioinformaticians to run bioinformatic tools with relative ease. Galaxy has an array of free tools, useful for many diverse purposes. One way the platform can be used is to determine evolutionary relationships between taxa utilizing genomic data to build phylogenies. [FASTTREE](#) is a tool designed to create maximum-likelihood phylogenies using methods such as maximum likelihood nearest-neighbor interchanges (NNIs) and minimum-evolution subtree-pruning-regrafting (SPR) (Price et al., 2010). FASTTREE is able to do this relatively quickly for massive sequence alignments (Price et al., 2010). [IQ-TREE](#) is another tool that builds maximum-likelihood trees (Nguyen et al., 2015). The IQ-TREE algorithm employs a combination of techniques which set it apart from other tools: perturbation of trees via hill-climbing NNI and utilizing a wide breadth of starting trees (Nguyen et al., 2015). It is expected that these tools will yield slightly different results with the same genomic data, as they each have different features that may be toggled before running the tool, and are each better fit for different uses.

Dataset

The NCBI database has a plethora of sequence data, including assemblies from over 4130 organisms (*NCBI Genome Data viewer*). Over the last few months (since June 2025) the following bee species' genomes have either been newly sequenced, and/or assembled and annotated, resulting in five entirely new or updated reference genomes (*NCBI Genome Data viewer*). Due to the updates to these genomes, the phylogenetic relationships between them would have only recently been assessed using this new data. Reference genomes tend to be more highly annotated, and for this reason the genes utilized in this tutorial (*Wnt10* and *Obp12*) were extracted from reference genomes. The *Wnt10* and *Obp12* gene sequences were concatenated into a single file (one file per species), so that resulting phylogenies were constructed using the combined data of two genes instead of one.

The following species' reference sequences were selected for this project:

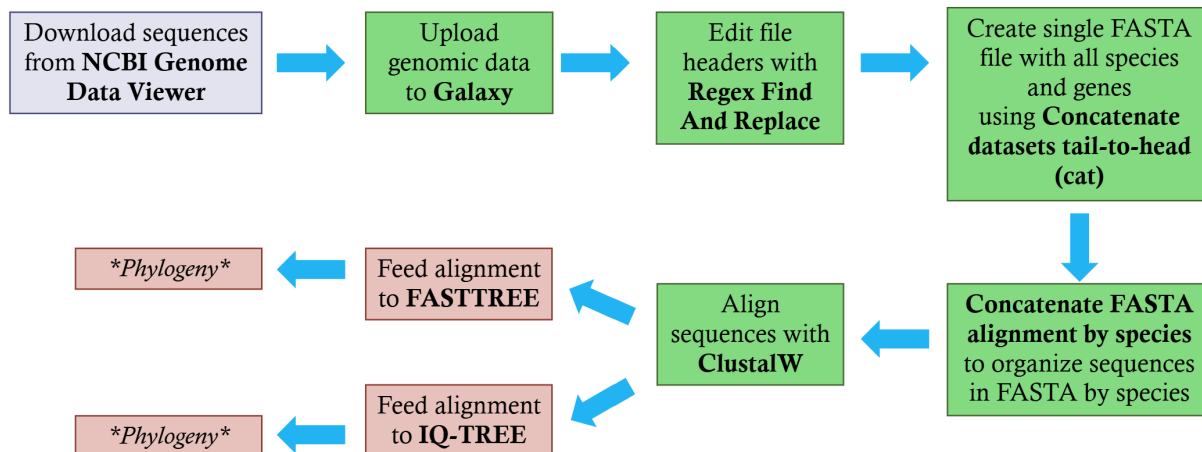
1. *Xylocopa sonora* (RefSeq: [GCF_050948175.1](#)),
2. *Andrena cerasifolii* (RefSeq: [GCF_050908995.1](#)),
3. *Halictus rubicundus* (RefSeq: [GCF_050948215.1](#)),

4. *Colletes latitarsis* (RefSeq: [GCF_051014445.1](#)).
5. *Bombus vancouverensis nearcticus* (RefSeq: [GCF_051014615.1](#)),
6. *Solenopsis invicta* (RefSeq: [GCF_016802725.1](#)), an ant species to be used as an outgroup.

The reason an ant species was chosen as an outgroup for this project is because the groups considered to be more “reliable” sister groups to bees (Pemphredoninae, Philanthinae, Ammoplanina) did not have reference genomes in NCBI. Like bees, ants belong to Hymenoptera and were even found to be more closely related to bees than most wasps are (Bailey, 2013)!

Large reference genomes can be difficult to manage and are likely to produce an error when attempting to align them in Galaxy. For this reason, a smaller portion of the genome was chosen to act as a representation of the species’ overall relatedness. Both a highly conserved gene, *Wnt10*, and a less conserved gene, *Obp12*, or odorant binding protein 12, were selected (Dearden et al., 2006; Johnson & Tsutsui, 2011). *Obp12* is a taxonomically restricted gene (TRG), meaning that it is unique to a given lineage, in this case, social insects such as bees and ants (Zielezinski et al., 2023). TRGs are useful for creating low-depth phylogenies, in other words, phylogenies comprised of closely related species (Zielezinski et al., 2013).

Protocol Overview



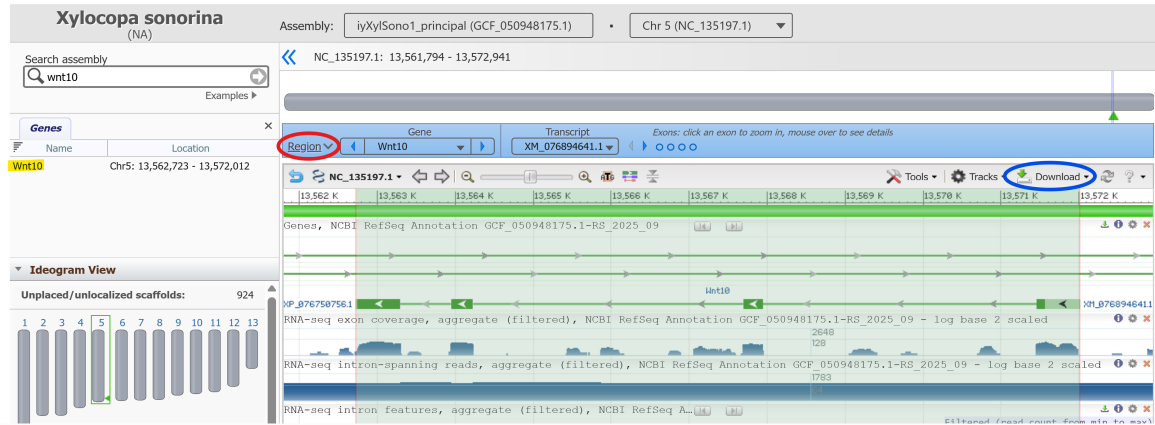
Protocol

1. Download sequences from [NCBI Genome Data Viewer](#)

- a. Once you have found your species of interest, search for each gene separately, and select from left-hand panel. In this case, **Wnt10** and *Obp12*.
- b. Click “Region” drop-down menu and choose “Select Gene”
- c. Download as FASTA file
 - i. Click “Download” drop-down menu

- ii. Click “Download FASTA”
- iii. Click “FASTA (All Selections)”

Genome Data Viewer



- d. In your computer, change the name of the downloaded file to something you will recognize as that species/ gene.
For example, “XSwt10”, or “ACobp12”

**Note: obp was assumed to be homologous to obp12 in Solenopsis invicta for the purposes of this manual, as this species does not have a gene specifically annotated as obp12.*

2. Navigate to <https://usegalaxy.org/>
3. **Upload** sequence data to Galaxy

**Note: you should upload two gene sequence files (obp12 and wnt10) for each of the six species; 12 different files should be uploaded to Galaxy in this step.*

- a. Click the “Upload” button in the top left corner of the Galaxy webpage



- b. Click the “Choose local file” button in the pop-up window, and select the files you wish to upload to Galaxy
 - i. These files will populate the “Drop files here” space in the pop-up window

- c. Select “Start”, then “Close”

Upload from Disk or Web to **Bioinformatics Project**

Regular Composite Collection Rule-based

Drop files here

Type (set all): Auto-detect Reference (set all): unspecified (?)

Choose local file Choose from repository Paste/Fetch data Start Pause Reset Close

- i. Your files will now appear in a column on the right-hand side of your screen (shown in yellow box)

The screenshot shows the Galaxy web interface. On the left is a sidebar with navigation options like Upload, Tools, Workflows, and Notifications. The main area displays a welcome message and a logo for the JXTX James P. Taylor Foundation. On the right, the 'History' panel for the 'Bioinformatics Project' is visible, showing a list of files. A yellow box highlights the following files in the list:

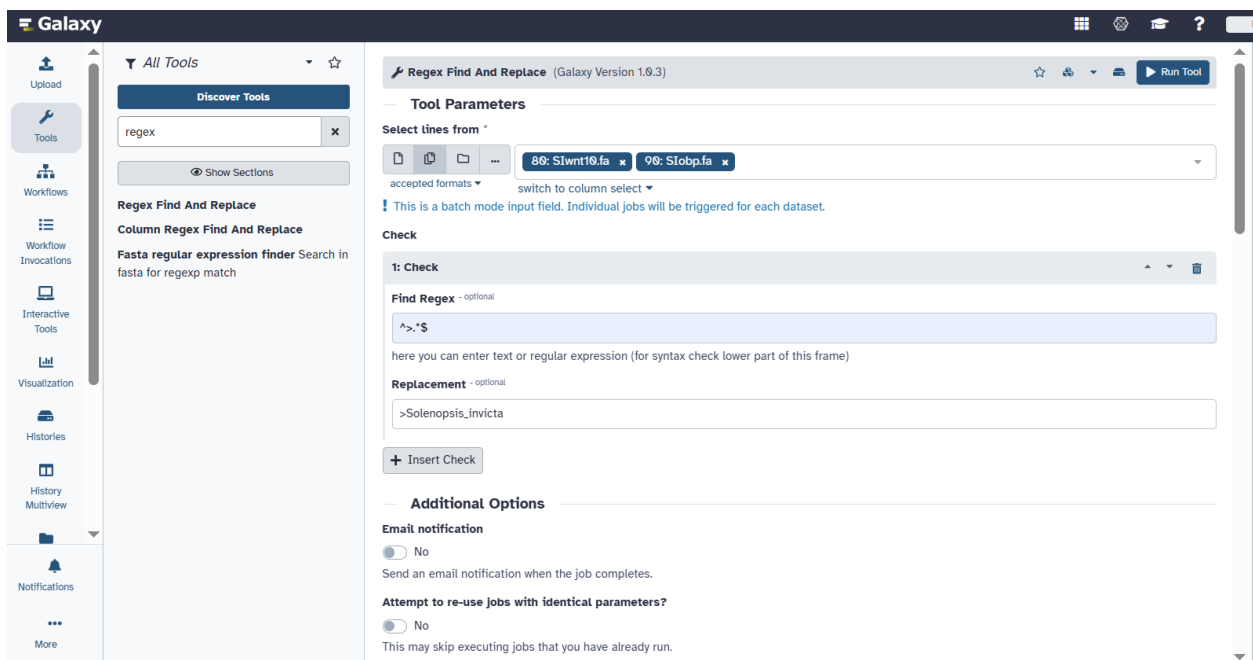
File Name	Size	Actions
95: XSobp12.fa	4.35 GB	Download, Edit, Delete
94: ACobp12.fa	56	Download, Edit, Delete
93: HRobp12.fa	40	Download, Edit, Delete
92: CLobp12.fa		Download, Edit, Delete
91: BVNobp12.fa		Download, Edit, Delete
90: SIobp.fa		Download, Edit, Delete
87: XS_E.fa		Download, Edit, Delete

4. Use the **Regex Find And Replace** tool to edit the header of each FASTA file

**Note: This is important for three reasons: (1) the header name will be used to label the branches, (2) Galaxy interprets the “|” symbol as an “or” in the default heading and will create errors downstream (Hiltemann, n.d.), (3) it is necessary that headers are in the correct format when concatenating by species, and are the same for each species in order to incorporate information from two different genes.*

- a. Using the “search tools” box, find the tool “Regex Find And Replace”
- b. Under the “Select lines from” heading, select the “Multiple datasets” icon: 
- c. Under this heading, click the grey arrowhead inside the textbox to the right, and select the files you wish to edit the header for
 - i. Do this one species at a time; choose the files for both genes of the same species
- d. Hit the “Insert Check” button
 - i. Under “Find Regex”, type: `^>.*$`
This tells Galaxy to replace the FASTA header from beginning to end (Hiltemann, n.d.)
 - ii. Under “Replacement”, type out the name you want instead of the default header. For example, `>Genus_species`

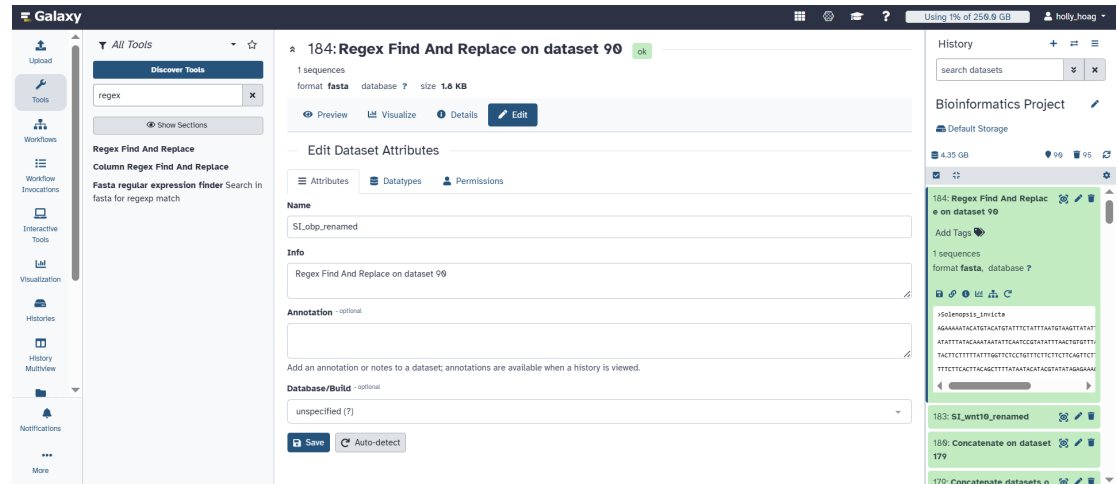
**Note: the “>” character is important to include, as it denotes the beginning of a header.*



The screenshot shows the Galaxy web interface. On the left is a sidebar with navigation options: Upload, Tools, Workflows, Workflow Invocations, Interactive Tools, Visualization, Histories, History Multiview, Notifications, and More. The main panel is titled 'Regex Find And Replace (Galaxy Version 1.0.3)'. It has a 'Tool Parameters' section with 'Select lines from' set to 'Multiple datasets' and two selected files: '80: Stwnt10.fa' and '90: Stobp.fa'. Below this is a 'Check' section with 'Find Regex' set to '^>.*\$' and 'Replacement' set to '>Solenopsis_invicta'. There is an 'Insert Check' button. At the bottom, there are 'Additional Options' including 'Email notification' (set to 'No') and 'Attempt to re-use jobs with identical parameters?' (set to 'No').

- e. Hit the “Run Tool” button near the top right of the screen.
Your output will appear in the same right-hand column as your other datasets
- f. *Optional:* You may change the name of the output files. This will make it easier to specify which file belongs to each gene and species, given the files of each species have the same header now.

View dataset (click the eye icon) > Edit > Attributes > Edit the “Name” field



- g. Repeat all of **Step 4, a-f** for the other five species.
- h. *Check*: at this point you should have 12 FASTA files across six species; each species should have two files, one for each gene. The header in each file should be the same within a species, while the title of each file may be different and indicate which gene the file represents.



etc.

5. Use **Concatenate datasets tail-to-head (cat)** in Galaxy to combine FASTA sequences from different species and different genes into a single FASTA file

**Note: Step five’s concatenation combines all 12 sequences (two gene sequences from each of the six species) into a single FASTA file. This is necessary since the concatenation tool in step six (which will combine the two gene sequences into a single sequence for each species) requires that all sequences of interest are within the same FASTA file.*

- a. Using the “search tools” box, find the tool “Concatenate datasets tail-to-head (cat)”
- b. Under the “Datasets to concatenate” heading, select the “Multiple datasets” icon:



- i. Under the same heading, click the grey arrowhead inside the textbox to the right. This will give a drop-down menu of all the datasets you have that are in a compatible format for this tool.
- ii. Click the checkbox for all datasets you wish to concatenate (in this case, both genes for all six species).

- c. Hit the “Run Tool” button near the top right of the screen.
- d. Your output will appear in the same right-hand column as your other datasets.
- i. Select the eye icon to view the concatenated dataset. You should see that this is a FASTA file with all datasets (12 total), separated by the headings for each sequence.

Note: there will be two sequences with the same heading for each species.

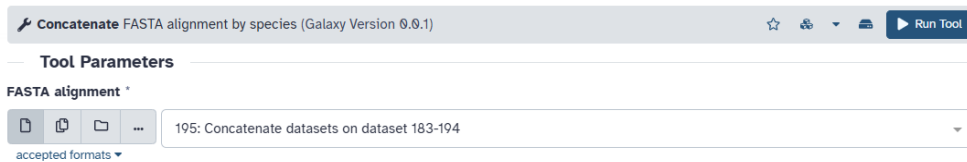
6. Use **Concatenate FASTA alignment by species** in Galaxy to combine FASTA sequences from the same species into a single sequence containing both genes

a. Using the “search tools” box, find the tool “Concatenate FASTA alignment by species”

b. Under the “FASTA alignment” heading, select the “Single dataset” icon: .

i. Under the same heading, click the grey arrowhead inside the textbox to the right. This will give a drop-down menu of all the datasets you have that are in a compatible format for this tool.

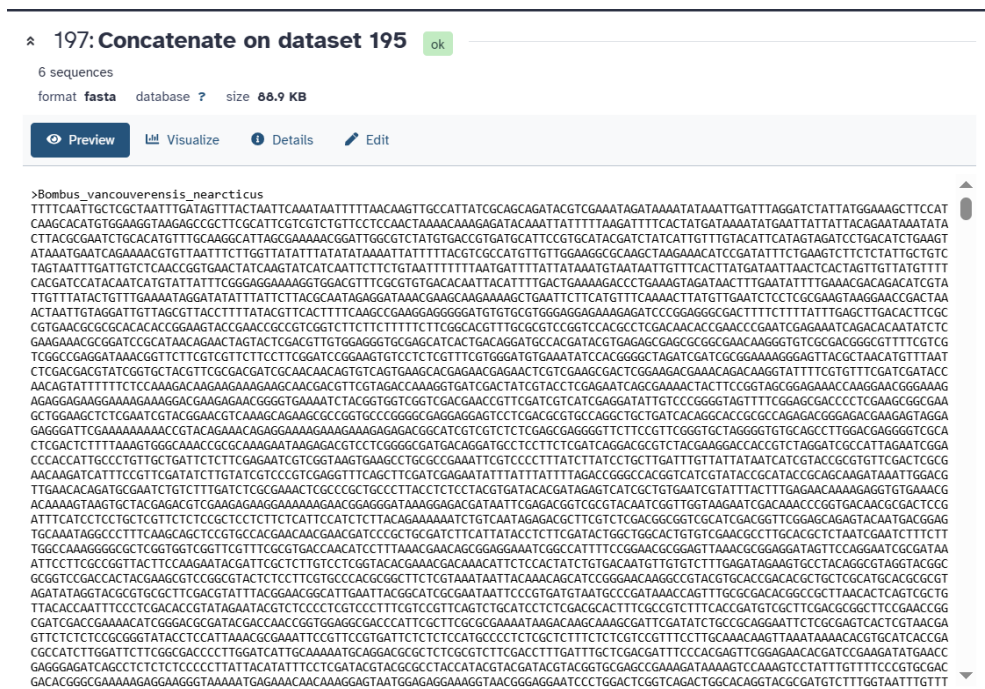
ii. Click the checkbox for the dataset you wish to concatenate (in this case, the concatenated file of all 12 sequences across six species)



c. Hit the “Run Tool” button near the top right of the screen.

d. Your output will appear in the same right-hand column as your other datasets.

i. Select the eye icon to view the concatenated dataset. You should see that this is a FASTA file with 6 sequences (combination of *Wnt10* and *Obp12*), one for each species.

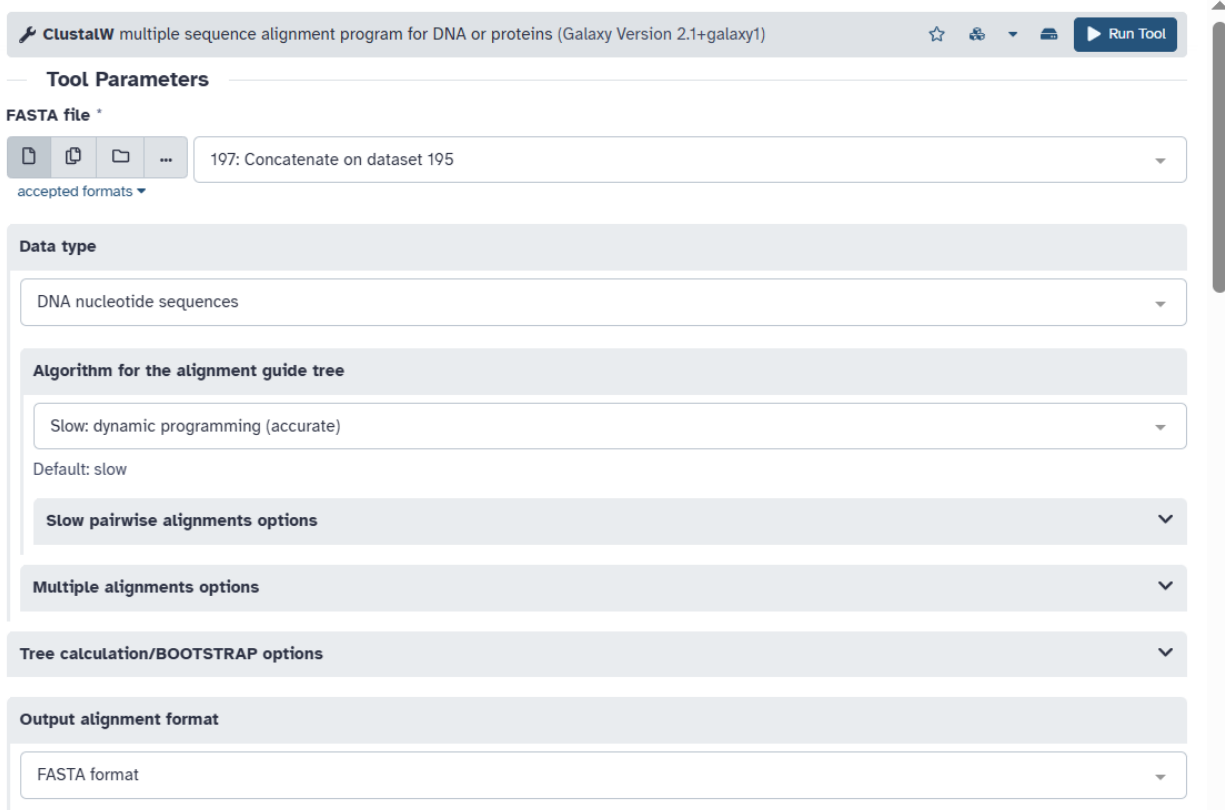


7. Use **ClustalW** in Galaxy to align sequences of all species of interest

The purpose of aligning sequences is so Galaxy can identify regions of similarity or dissimilarity between species' genomic data, which will be used to create the phylogeny.

**Note: though ClustalW is used in this tutorial, you may use any multiple sequence DNA alignment tool you like, as long as the output file is in FASTA format.*

- Using the “search tools” box, find the tool “ClustalW”
- Under the “FASTA File” heading, select the “Single dataset” icon: .
- Under the same heading, click the grey arrowhead inside the textbox to the right, and select your concatenated file.
- Change “Output alignment format” to “FASTA format”.
- Ensure that “Data type” is set to “DNA nucleotide sequences” and keep all unmentioned settings as the default.



ClustalW multiple sequence alignment program for DNA or proteins (Galaxy Version 2.1+galaxy1) ☆ 📄 ⌵ 📁 ▶ Run Tool

Tool Parameters

FASTA file *

   ... 197: Concatenate on dataset 195 ⌵

accepted formats ▼

Data type

DNA nucleotide sequences ⌵

Algorithm for the alignment guide tree

Slow: dynamic programming (accurate) ⌵

Default: slow

Slow pairwise alignments options ▼

Multiple alignments options ▼

Tree calculation/BOOTSTRAP options ▼

Output alignment format

FASTA format ⌵

- Hit the “Run Tool” button.
This gives two outputs: dnd (.nhx alignment file) and FASTA (FASTA alignment file).

Now that you have a FASTA file with the data from both genes aligned for all species of interest, the FASTA file can be used as the input for both FASTTREE and IQ-Tree.

[Click here to view the resulting FASTA file](#)
[\[sequence alignment for all species\].](#)

Plug your FASTA file into **FASTTREE**

1. Using the “search tools” box, find the tool “FASTTREE” and click it
2. Under the “Aligned sequences file (FASTA or Phylip format).” heading, ensure the correct file type is entered. In this case, “fasta”
3. Under the “FASTA file” heading, select the “Single dataset” icon: .
4. Under the same heading, click the grey arrowhead inside the textbox to the right, and select your ClustalW FASTA file.
5. Scroll down to the heading “Protein or nucleotide alignment” and select “Nucleotide” from the drop-down menu
6. Keep all other settings as their default

The screenshot shows the Galaxy web interface with the FASTTREE tool configuration page. The tool parameters are set to 'fasta' format, '198: ClustalW on dataset 197: fasta' as the FASTA file, and 'Nucleotide' for the alignment type. The 'Run Tool' button is visible in the top right of the tool panel.

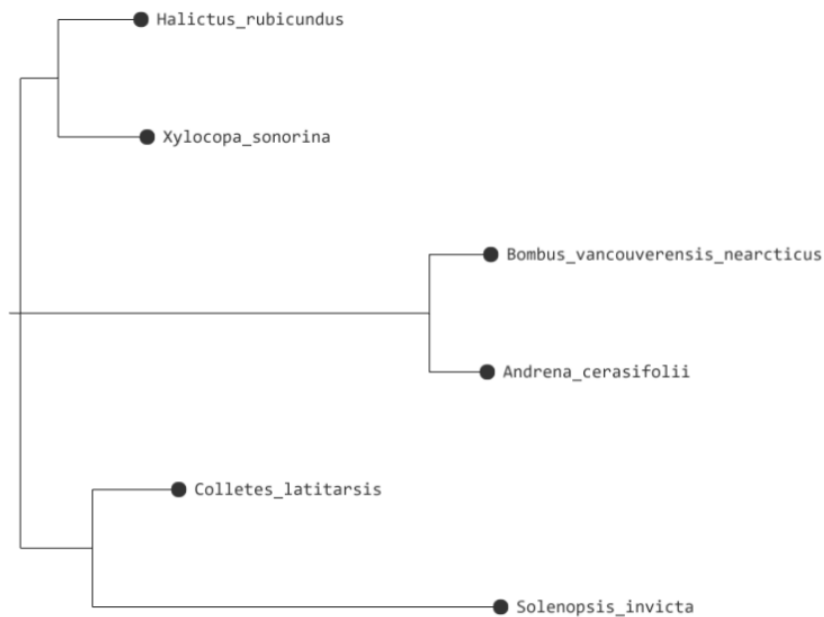
7. Hit “Run Tool”

You will be given 1 output - .nhx file

8. Tree can be visualized directly in Galaxy.

- Select the “View” icon of your output file
- Navigate to the “Visualize” tab
- Select “Phylogenetic Tree”, and your resulting tree will populate your screen

The screenshot shows the Galaxy visualization interface for the output file '205: FASTTREE on dataset 198:tree.nhx'. The file is in 'nhx' format, 262 b in size, and has 1 line. The 'Visualize' tab is selected, showing a search bar and a list of visualization options including 'Phylogenetic Tree'.



PhyloCanvas

9. *Optional:* Tree may also be copied from Galaxy as .nhx (extended Newick) file and uploaded to iTOL (below) for a more customizable platform (Letunic, *Interactive tree of life*).

**Note: an extended Newick file is a specifically formatted string of characters which communicates the structure of a phylogenetic tree in text format. It may be fed to a phylogeny visualizer in order to view a given tree.*

- a. Copy the text from the “Preview” tab of your .nhx output

Galaxy

205: FASTTREE on dataset 198:tree.nhx

1 line
format **nhx** database ? size **262 b**

FASTTREE build maximum-likelihood phylogenetic trees

qime2 phylogeny align-to-tree-mafft-fasttree Build a phylogenetic tree using fasttree and mafft alignment

qime2 phylogeny fasttree Construct a phylogenetic tree with FastTree.

```
((Halictus_rubicundus:0.212485487,Xylocopa_sonorina:0.228146489)0.999:0.096569839,
(Bombus_vancouverensis_nearcticus:0.127482023,Andrena_cerasifolia:0.148425177)1.080:1.047321557,
(Colletes_lutitarsis:0.228788387,Solenopsis_invicta:1.045438859)1.000:0.184665926);
```

History

search datasets

Bioinformatics Project

Default Storage

4.35 GB

205: FASTTREE on dataset 1 98:tree.nhx

Add Tags

1 line
format **nhx** database ?

FastTree Version 2.1.10 Double precision (No SSE3)
Alignment: standard input

((Halictus_rubicundus:0.212485487,Xylocopa_sonorina:0.228146489)0.999:0.096569839,
(Bombus_vancouverensis_nearcticus:0.127482023,Andrena_cerasifolia:0.148425177)1.080:1.047321557,
(Colletes_lutitarsis:0.228788387,Solenopsis_invicta:1.045438859)1.000:0.184665926);

204: IQ-TREE on dataset 19 8: Report and Final Tree

203: IQ-TREE on dataset 19 8: Occurrence Frequencies in

- Navigate to the [iTOL upload tab](#)
- Paste your copied text into the “Tree Text” box and select “Upload”.
Your screen will populate with your resulting tree, along with a side-menu for customization options.

iTOL INTERACTIVE TREE OF LIFE

Tree of Life Upload Data sharing Help

Sign in Register

Tree scale: 0.1

Solenopsis invicta
Colletes latitarsis
Andrena cerasifolia
Bombus vancouverensis nearcticus
Xylocopa sonorina
Halictus rubicundus

Control panel

Basic Advanced Datasets Export

Mode Circular Rectangular Unrooted

Mode options

Rotation: 0°

Branch lengths: Use Ignore

Invert tree: Yes No

Slanted: Yes No

Labels

Display Hide

Label options

Font: Arial

Font style: 20px B I

Position: Aligned At tips

Alignment: Left Right

Rotation: On Off

Shift: 0px

Branch options

Line style: 1px

Color gradient: On Off

Dashed lines: 0.3px

Tree views Undo Reset tree

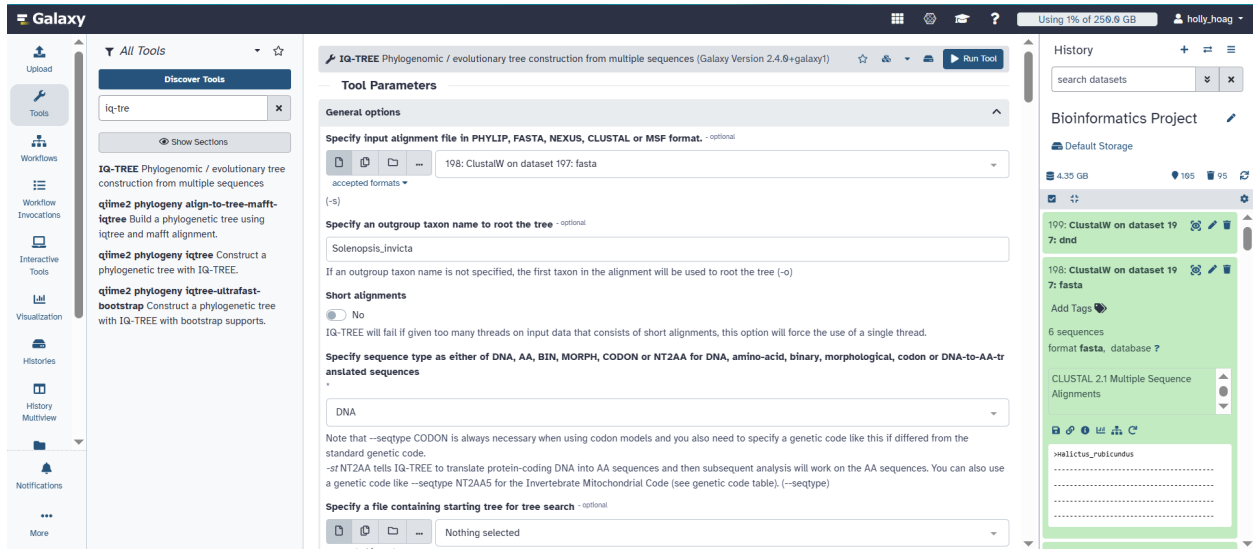
Plug your file into IQ-TREE

- Using the “search tools” box, find the tool “IQ-TREE” and click it
- Under the “Specify input alignment file in PHYLIP, FASTA, NEXUS, CLUSTAL or MSF format.” heading, select the “Single dataset” icon: .
- Under the same heading, click the grey arrowhead inside the textbox to the right, and select your ClustalW FASTA file.

- Under the “Specify an outgroup taxon name to root the tree” heading, enter the header of your outgroup species, but exclude the “<” symbol.

Note: if you skip this step, IQ-TREE will automatically select the first species in your alignment file to be the outgroup.

- Ensure sequence type field matches the type of data you have. In this case, “DNA”. Keep all other settings as their default.



- Hit “Run Tool”

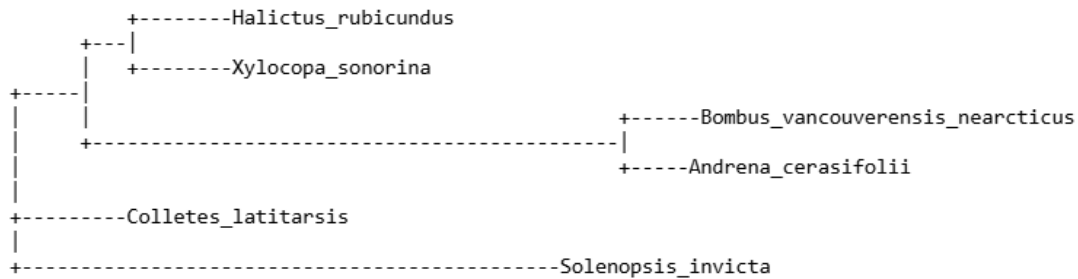
You will be given 5 outputs:

BIONJ Tree
MaxLikelihood Tree
MaxLikelihood Distance Matrix
Occurrence Frequencies in Bootstrap Trees
Report and Final Tree

The respective outputs and file types are as follows:

- BIONJ Tree- .nhx
- MaxLikelihood Tree- .nhx
- MaxLikelihood Distance Matrix- .mldist
- Occurrence Frequencies in Bootstrap Trees- .nex
- Report and Final Tree- .iqtree

7. Select “Report and final tree” and scroll down to see the resulting phylogeny



8. *Optional:* Tree may also be visualized in iTOL as was shown in the FASTTREE instructions (step 9 above). For step 9a, the newick format for your IQ-TREE output will be found just below your resulting phylogeny.

Tree in newick format:

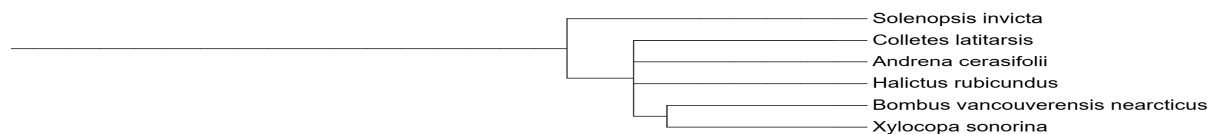
```

(((Halictus_rubicundus:0.2263542877,Xylocopa_sonorina:0.2342289947):0.1025322193,
(Bombus_vancouverensis_nearcticus:0.1726544797,Andrena_cerasifolii:0.1498318182):1.1257941331):0.1584382497,Colletes_latitarsis:0.
2562415909,Solenopsis_invicta:1.1484883341);
  
```

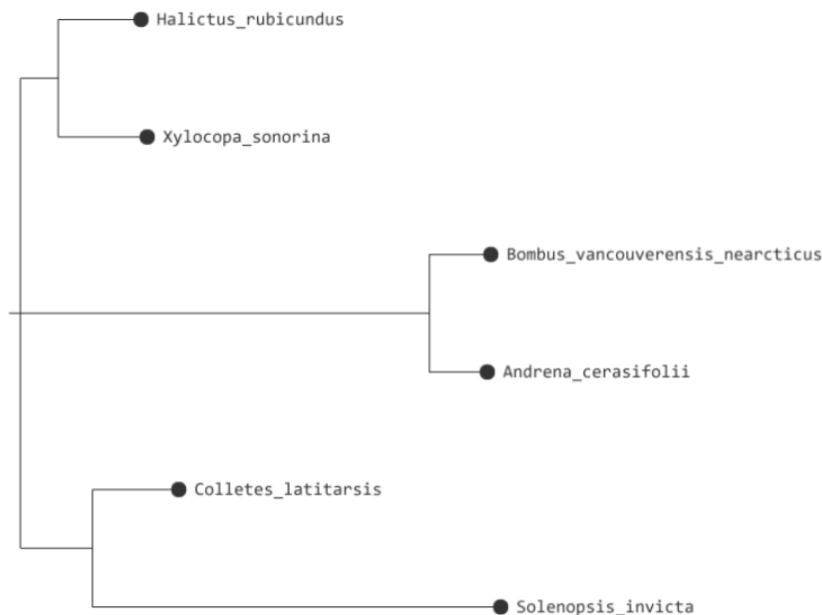
Discussion and Conclusion

Comparison of Trees

The following tree was created using phyloTv2, a program that generates a file for a phylogenetic tree, which must then be fed to iTOL for visualization. It is provided here as a reference which incorporates NCBI data and constructs a tree representing what is currently understood to be the taxonomic relationship between species, i.e. how these species are organized in the NCBI taxonomic database (NCBI, n.d.; Letunic, 2015).

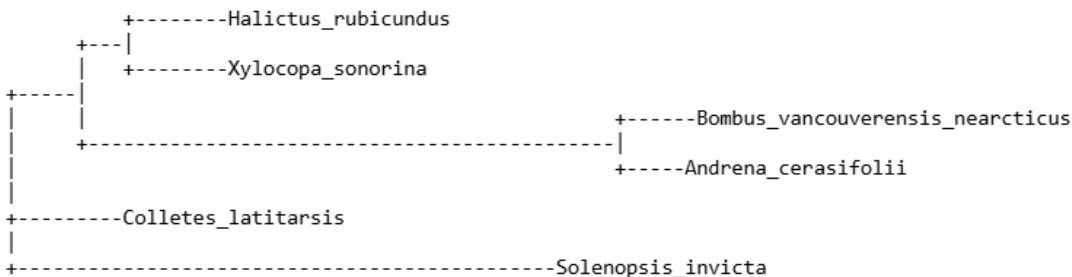


Above tree generated with phyloT v2 using NCBI taxonomy. Visualized with iTOL.

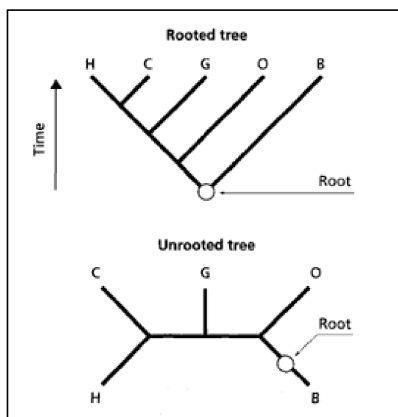


PhyloCanvas

Above tree generated with FASTTREE and visualized within the “Phylogenetic Tree” feature within Galaxy.



Above tree generated with IQ-TREE. Tree shown is from the Report and Final Tree output.



The phylogenies produced by FASTTREE and IQ-TREE share some meaningful information; the algorithms chose the same species pairs for showing the greatest degree of relatedness. On both phylogenies, *Bombus vancouverensis nearcticus* and *Andrena cerasifolii* as well as *Xylocopa sonorina* and *Halictus rubicundus* were selected as being more closely related to each other than to any other taxa. The IQ-TREE phylogeny had *Solenopsis invicta* manually selected as an outgroup, though the algorithm determined that *Colletes latitarsis* was an equally

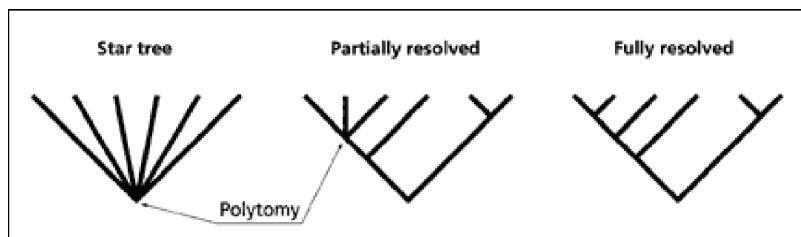
distant group as compared to the other taxa, or the tree's "ingroup". For this reason, both *C. latitarsis* and *S. invicta* appear as outgroups in IQ-TREE. The FASTTREE algorithm sets *C. latitarsis* and *S. invicta* as an additional pairing with a high degree of relatedness, instead of as an outgroup to the other taxa in the phylogeny. In other words, the FASTTREE output is *unrooted* and has no outgroup (see the above diagram from Page & Holmes (2009)) where a rooted versus unrooted tree of the same taxa is compared (the circle placed on the unrooted tree represents where the root *would be* if the phylogeny was rooted). It is worth noting that additional tools, such as iTOL, can be used to manually add a root to a FASTTREE output, should the user wish to.

When compared to the NCBI taxonomy tree, the FASTTREE and IQ-TREE outputs are quite different. While *S. invicta* is the outgroup, the remainder of the tree does not communicate much about the relatedness between species; the only additional information this tree provides is that *X. sonora* and *B. vancouverensis nearcticus* are more closely related to each other than to other taxa. This conclusion does not align with the results from IQ-TREE and FASTTREE, and can be attributed to the fact that the Galaxy tools take only two genes (*Wnt10* and *Obp12*) into account when creating the phylogenies, whereas the NCBI taxonomy represents the currently accepted evolutionary relationship based on a much wider range of data such as “published literature, web-based databases, and the advice of sequence submitters and outside taxonomy experts” (NCBI, n.d.). All in all, IQ-TREE produces an output more similar to that of NCBI taxonomy.

Analysis of Tools

The two Galaxy-based tools of FASTTREE and IQ-Tree created separate phylogenies, based on the same genome data, that are not entirely similar.

According to [Galaxy](#), for the resulting FASTTREE phylogeny, “The placement of the root is not biologically meaningful[...] If all sequences are unique, then the tree will be fully resolved (the



root will have three children and other internal nodes will have two children)” (*Galaxy*). This means that the FASTTREE algorithm aims to follow a “template” for placing species on branches; for example, if you input 6 species into FASTTREE, it is given that the tree will have three “primary” branches, each with two “secondary” branches (as can be seen in the above FASTTREE output). This is what it means for a phylogeny to be “fully resolved”- each node is “bifurcated” and has no more than two branches coming out of it (Page & Holmes, 2009). When using FASTTREE, this will be the case unless two sequences are identical (*Galaxy*). If two sequences are identical, the resulting tree will have a “polytomy”, meaning more than two branches originating from a single internal node (Page & Holmes, 2009). See the above diagram

from Page & Holmes (2009) for a visual representation of polytomy and resolved phylogenies. IQ-TREE will create a phylogeny whose shape is deemed most representative of the relationships between sequences. In other words, an IQ-TREE phylogeny is not constrained to a template in the way that a FASTTREE phylogeny is. This means it is possible for IQ-TREE to create an unresolved or partially resolved phylogeny.

In IQ-TREE the root (or, the automatically selected outgroup) is not necessarily representative of biological reality; the root selection is not based on sequence similarity data. However, in IQ-TREE the user has the option to manually set the root, but this requires that the user has knowledge of which species represents the true outgroup. This is in contrast to FASTTREE where a root *cannot* be manually set within Galaxy, meaning that there is no discernable outgroup for these trees.

With all items considered, FASTTREE may be more appropriate for gaining insight into approximate relationships for large groups of taxa based on genomic data; while FASTTREE tends to produce a less accurate phylogeny, it creates these trees relatively quickly for large datasets (Zhou et al., 2018). When debating between the usage of FASTTREE or IQ-TREE, the user should consider which parameters are more important to them- speed (FASTTREE) or accuracy (IQ-TREE).

Users should be aware of a few potential pitfalls of this protocol. When choosing which gene(s) will be used to create the phylogenies, it is important to note the following:

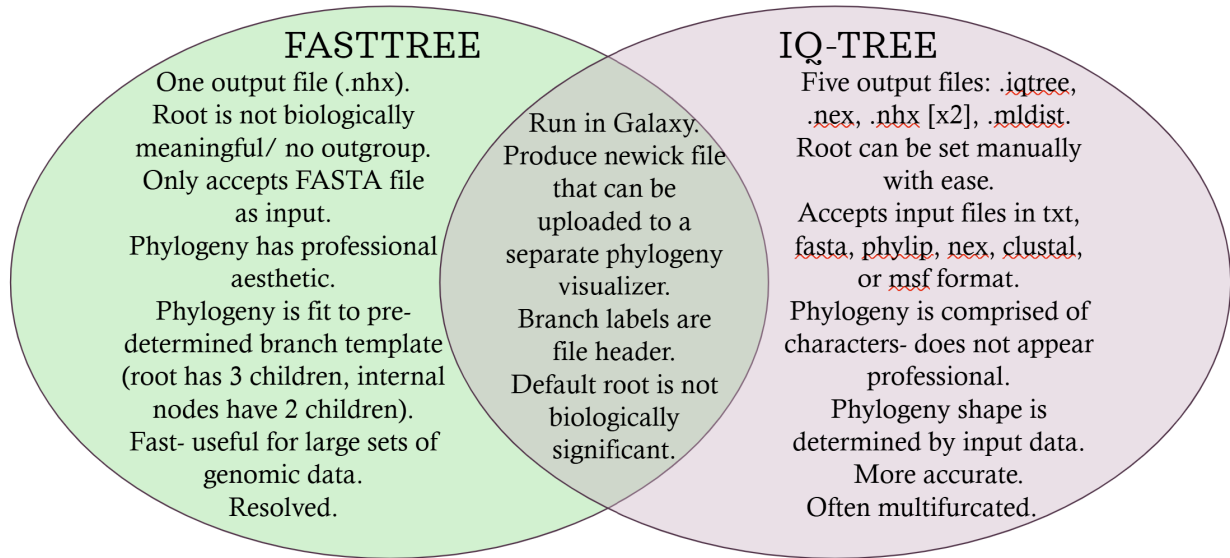
- The same gene may appear multiple times in the genome. For the purpose of simplicity in this manual, only one locus was downloaded for analysis (though, it should be noted that most species had only one locus for a given gene).
- Homologous genes may have different names between species and therefore may have been incorrectly excluded or included in the dataset used to create the trees.
 - For example, the ant species, *Solenopsis invicta*, had a gene named “Obp”, which was assumed to be homologous to “Obp12” from the bee species analyzed.

Therefore, when creating a phylogeny, it is of great importance to ensure that the data chosen to construct the tree is (theoretically) representative of the true evolutionary relationships.

Conclusion

Both FASTTREE and IQ-Tree are easy ways to create phylogenies based on genome sequence data. Although fewer steps are required to get a professional-appearing phylogeny from FASTTREE, IQ-tree is a more customizable way to build a phylogeny within Galaxy, with outputs that are more likely to represent biological reality based on root placement and overall phylogeny structure. While FASTTREE may be better for quickly creating a phylogeny from a large set of

genomic data, IQ-TREE is likely to produce more accurate results.



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**Note: some of these references were provided by ChatGPT. Information was manually confirmed before incorporation in this manual.*

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